

CHANGE REQUEST

23.288 CR 1327 rev 1 Current version: 19.1.0

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Proposed change affects: UICC apps ☐ ME ☐ Radio Access Network ☐ Core Network ☒

Title:	Support of VFL model training and inference with client intermediate results sharing between VFL clients	
Source to WG:	Samsung [Nokia, Apple, ZTE, Lenovo, OPPO]	
Source to TSG:	SA2	
Work item code:	AIML_CN	Date: 2025-02-10
Category:	B Use <u>one</u> of the following categories: F (correction) A (mirror corresponding to a change in an earlier release) B (addition of feature), C (functional modification of feature) D (editorial modification) Detailed explanations of the above categories can be found in 3GPP TR 21.900 .	Release: Rel-19 Use <u>one</u> of the following releases: Rel-8 (Release 8) Rel-9 (Release 9) Rel-10 (Release 10) Rel-11 (Release 11) ... Rel-15 (Release 15) Rel-16 (Release 16) Rel-17 (Release 17) Rel-18 (Release 18)

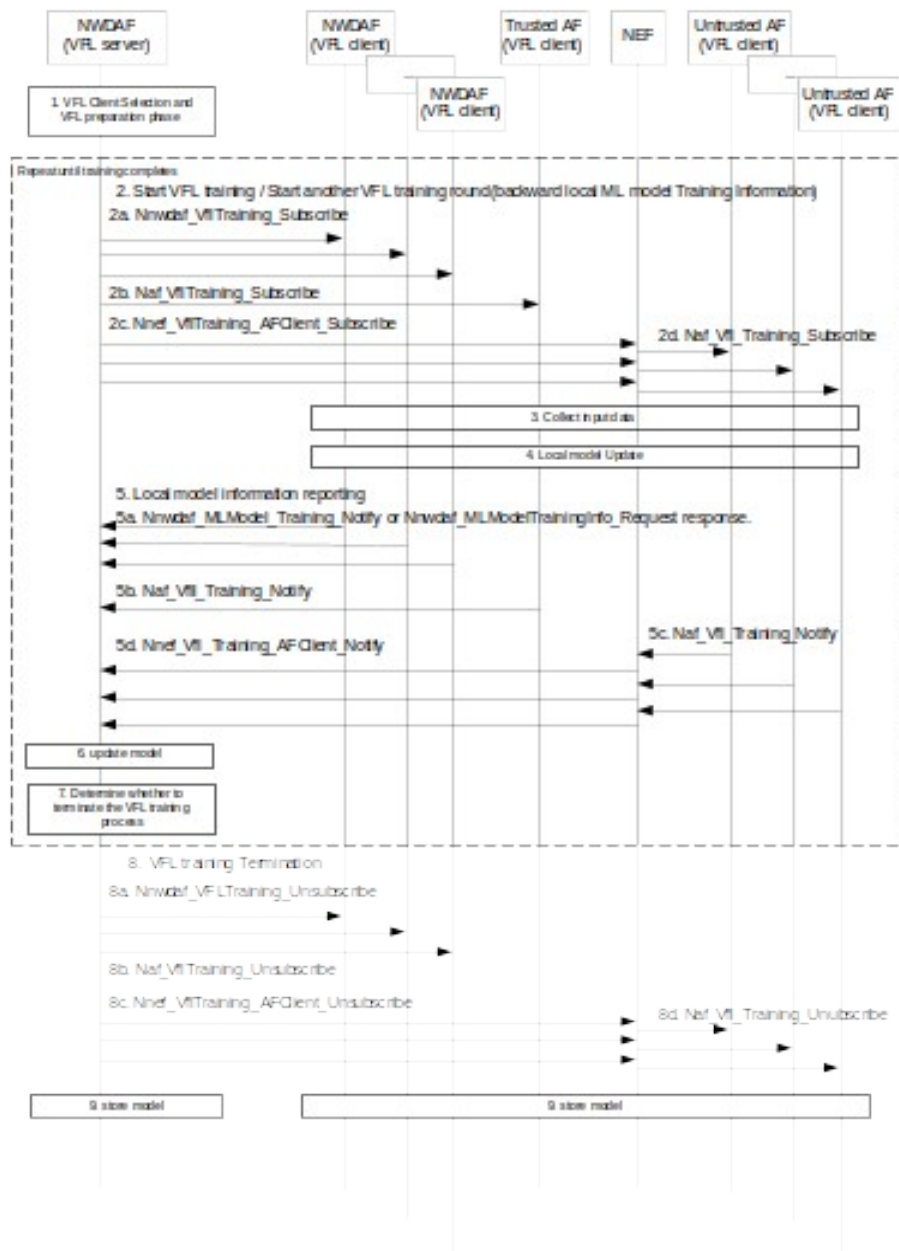
Reason for change:	Rev1: <u>Based on discussion in S2-2501347, when VFL using split models, whether the client intermediate results can be aggregated is based on gradient. As SA2 has agreed, the VFL models are pre-configured and the supporting gradient information of VFL client are indicated (implicitly and explicitly) by VFL interoperability indicator(s) and VFL interoperability information used during discovery and preparation stage. Therefore, the existing discovery and preparation framework is sufficient enough to support the VFL server to choose VFL client aggregator and indirect VFL clients. No VFL model is to be shared in 3GPP standardised approach.</u> <u>There is an EN in 0232r05:</u> <u>Editor's Note: It is FFS how the client aggregator is selected and may not be selected/configured by VFL server, and also if this capability needs to be registered into NRE</u> <u>Different from VFL server, the VFL client aggregator does not have VFL labels. It only aggregates the client intermediate results and then compute its own intermediate result. Therefore, the VFL client aggregator and indirect VFL clients still align the definition of VFL client. No additional capability is required.</u> <u>The simplest way is to allow the VFL client aggregator and indirect VFL clients can be selected by VFL server based on the interoperability information received during discovery and/or preparation, as the VFL clients do not share interoperability information in current framework.</u> <u>Step 2 of clause 6.2H.2.4.1 documented a scenario that no VFL client is selected for VFL inference. However, VFL requires the coordination between VFL clients and VFL server for VFL training and inference. The above scenario is not support in VFL; and therefore, should be removed.</u>
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	Rev0: The conclusions for VFL training documented in clause 8.2 of TR 23.700-84, including: <i>P#2.4.5: VFL clients may also provide intermediate results (e.g. gradient information, loss information) to other VFL clients as instructed by the VFL server.</i> As it has been agreed in expection sheet (in SP-241513/ S2-2413037), the following issue should be addressed: <i>Vertical Federated Learning (VFL):</i> <i>Whether and how to exchange intermediate training results between VFL clients during VFL training and inference</i> As discussed in DP S2-2500233, VFL algorithms using split neutral network (NN) is widely used which can reduce the energy consumption and improve the overall VFL efficiency. When using the split neutral network (NN) network, the VFL clients may share client intermediate results with a selected VFL client during the model training and inference for intermediate results aggregation to distribute the computation load of VFL server. Furthermore, when the VFL server is untrusted AF, exposing the client intermediate results of every single VFL client may increase the risks of 5GC privacy issues, as no privacy preserving methods are specified. Therefore, support client intermediate results sharing between VFL clients is will beneficial for R19 AIML_CN VFL operation. To simplify 3GPP work, in this release, VFL model training and inference with VFL client information sharing will be only support among NWDAF VFL clients; the (untrusted) AF VFL clients are not considered. This CR specifies the procedures for VFL model training with client intermediate results sharing between VFL clients. The ENs on VFL client information sharing during VFL model and inference are removed.										
Summary of change:	Rev1: - Minor clarification on VFL client information sharing. - Remove invalid VFL scenario from Step 2 of clause 6.2H.2.4.1. - Other Rev0: Introduce new steps to support VFL model training and inference with VFL client information sharing on top of the agreed procedures. Remove ENs relevant to VFL client information sharing.										
Consequences if not approved:	Conclusions for FS_AIML_CN KI#2 or VFL cannot be fully supported by 5GC.										
Clauses affected:	6.2H.2.3.1, 6.2H.2.3.2, 6.2H.2.4.1, 6.2H.2.4.2										
Other specs affected: (show related CRs)	<table><tr><th>Y</th><th>N</th></tr><tr><td></td><td>X</td></tr><tr><td></td><td>X</td></tr><tr><td></td><td>X</td></tr></table>	Y	N		X		X		X	Other core specifications	TS/TR ... CR ...
Y	N										
	X										
	X										
	X										
		Test specifications	TS/TR ... CR ...								
		O&M Specifications	TS/TR ... CR ...								
Other comments:											
This CR's revision history:											

*****Start of Changes*****

6.2H.2.3.1 Training Procedure for Vertical Federated Learning when NWDAF is acting as VFL server

The figure 6.2H.2.3.1-1 below shows the training procedure for Vertical Federated Learning when NWDAF is acting as VFL server.



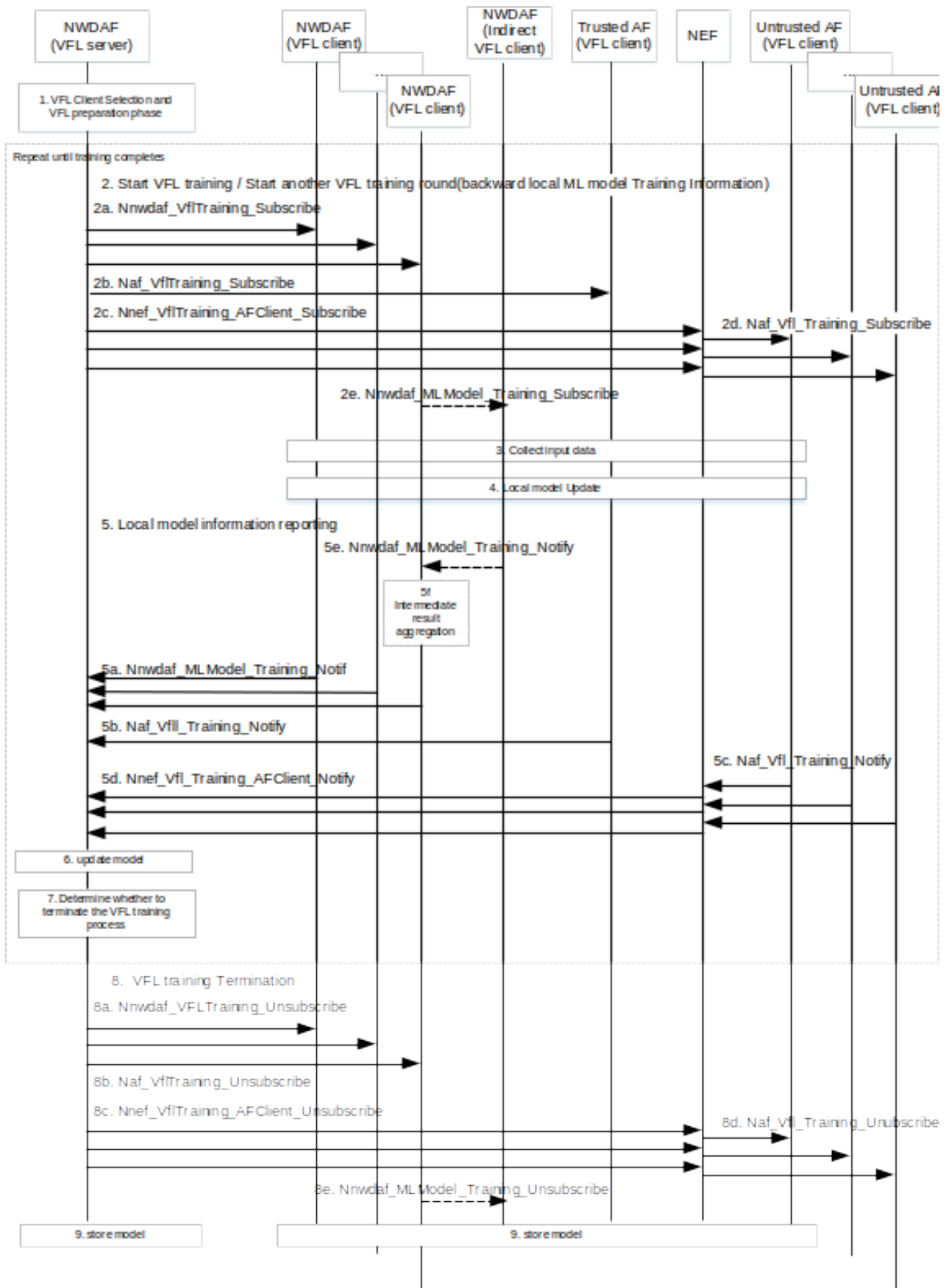


Figure 6.2H.2.3.1-1: Training procedure for Vertical Federated Learning when NWDAF is acting as VFL server

Editor's note: How the NEF assists the VFL training process as well as whether the service operations going via NEF is using the existing or new service operation are FFS.

Editor's note: The details of the services in the procedure and whether VFL Training Start Flag is needed are FFS.

Editor's note: It is FFS whether sample/feature information is required to be provided or updated in each training.

Editor's note: Whether and how to include interoperability information in the VFL training procedure is FFS.

Editor's note: Whether and how to define the trigger of VFL training is FFS.

Editor's note: Whether and how to transfer the confirmation in VFL Preparation Phase at the beginning of VFL Training Phase is FFS.

Editor's Note: It is FFS how the client aggregator is selected and may not be selected/configured by VFL server, and also if this capability needs to be registered into NRF.

1. The NWDAF acting as VFL server determines the VFL clients that participate in VFL procedure in the VFL clients discovery and preparation phase as described in the clause 6.2H.2.1 and clause 6.2H.2.2.

NOTE 1: VFL Server can determine to start the training based on local configuration and agreement among vendors and/or application providers participating in the same group for specific VFL task(s).

Steps 2-6 are repeated until the training termination condition is reached.

2. To start the VFL training, the VFL server sends a request to start the training to all selected VFL clients. The request includes VFL correlation ID, at least the parameters negotiated during the preparation phase. Optionally, the VFL Server includes: Analytic filter information, maximum response time (i.e. the maximum time between VFL clients receive intermediate model training information and send back intermediate training result).

Editor's note: Whether the parameters negotiated in the preparation phase are provided at the end of the preparation phase or at the start of the training is FFS.

If the VFL procedure continues in subsequent iterations, the VFL server sends a request for a new VFL training iteration containing the VFL correlation ID and intermediate model training information to each of the VFL clients for next round of VFL training.

2a. The VFL server sends a Nnwdaf_VFLTraining_Subscribe to the selected NWDAF VFL clients(s).

2b. The VFL server sends a Naf_VFLTraining_Subscribe to the selected trusted AF VFL clients(s).

2c. For each selected untrusted AF VFL clients, the VFL server sends a Nnef_VFLTraining_AFClient_Subscribe to the NEF handling that AF.

2d. For each selected untrusted AF VFL clients, the NEF sends a Naf_VFLTraining_Subscribe to that AF. The NEF may also translate the analytic filter information if needed, e.g. TAIs into geographical area.

2e. If a NWDAF VFL client is selected as VFL client aggregator by the VFL server, this AF NWDAF VFL client may send Nnwdaf_VFLTraining_Subscribe to the other one or more indirect NWDAF VFL client(s) as configured by the VFL server from which it desires to receive the client intermediate training results.

NOTE 2: In this release, the same NF associated with a VFL Server or VFL Client capability during the VFL training for a VFL correlation ID is also the same NF during the VFL inference.

Editor's note: Additional Parameters to be provided in the request are FFS.

Editor's note: It is FFS whether and how the local ML model is obtained by VFL Client in VFL training process.

3. [Optional] Each VFL client collects its local data by using the current mechanism if the VFL client has no local data already available. The data used by each VFL Client is collected as per alignment information.
4. During VFL training procedure, each VFL client further trains the local ML model associated with the same VFL Correlation ID based on their own collected or available data and when applicable (e.g. after the first round of training) and possible intermediate model training information distributed by the VFL server in the previous

training iteration. Each VFL Client computes and reports the client intermediate training result of the local ML model to the VFL server.

NOTE 3: The intermediate model training information and intermediate training result are constructed in per sample granularity.

Editor's note: It is FFS and may depend on the service design: When the clients report the client intermediate training result, it also includes the corresponding VFL correlation ID.

~~Editor's note: Whether and how to exchange intermediate training result among VFL Clients is FFS.~~

5. Each VFL client reports the computed client intermediate training result of the local ML model to the VFL server.

5a. A NWDAF VFL client sends a Nnwdaf_VFLTraining_Notify.

5b. A trusted AF VFL client sends a Naf_VFLTraining_Notify to the VFL server.

5c. An untrusted AF VFL client sends a Naf_VFLTraining_Notify to the NEF.

5d. For each untrusted AF VFL client, the NEF converts any external identifiers to internal identifiers and sends a Nnef_VFLTraining_AFClient_Notify to the VFL server.

5e – 5f. An indirect NWDAF VFL client may send the client intermediate training results to the NWDAF VFL client aggregator from which it received the subscription request step 2e. This NWDAF VFL client aggregator aggregates the received client intermediate training results from the indirect NWDAF VFL clients, performs local computation, then sends one Nnwdaf_VFLTraining_Notify that includes its client intermediate training result to the VFL server.

6. The VFL server may collect the local data and generate its own local intermediate training result. The NWDAF acting as VFL Server computes the intermediate model training information (e.g. gradient information or loss information) based on the VFL Client(s) intermediate training result(s) received in step 4, its own local intermediate results and the label. The intermediate model training information is used for updating the models of VFL clients. Different intermediate model training information may be computed for different VFL clients and for the VFL Server itself.

The VFL server may also compute the ML model metric (e.g. ML model accuracy) based on all the intermediate training result received from VFL clients and the label.

Editor's note: Whether weight of the VFL Client is computed by VFL server is FFS.

Editor's note: Whether VFL server and VFL clients share feature information is FFS.

7. [Optional] The NWDAF acting as VFL server evaluates (e.g. based on the convergence of a loss function or loss value and/or if the pre-set iteration number is reached) whether VFL Training process converged. If the VFL Server evaluates the VFL Training process did not converge, the NWDAF acting as a VFL Server determines another round of VFL training is required and repeats step 2 - 6. If the VFL Server evaluates the VFL training process converged, it determines the VFL Training is completed. In this case, the VFL Server terminates the current VFL training process via step 7.

The VFL training termination decision may be also made as follows:

Based on the consumer request, the VFL server sends VFL status report to the consumer. The status report may include model metric (e.g. ML model accuracy).

Editor's note: The content of the VFL status report is FFS.

Editor's note: Whether VFL server sending convergence report to the VFL client and what is convergence report are FFS.

The consumer decides whether the current model can fulfil the requirement, e.g. ML model metric is satisfactory for the consumer and determines to stop or continue the training process. The consumer continues the training process or stops the training process.

Based on the subscription request sent from the consumer, the VFL server updates or terminates the current VFL training process.

Editor's note: Whether the ML model metric (e.g. ML model accuracy) defined for HFL can be re-applied to VFL is FFS.

8. The VFL server sends VFL training termination message to VFL Client if it decides to terminate the VFL training process, the termination message contains VFL Correlation ID.

8a. The VFL server sends a Nnwdaf_VFLTraining_Unsubscribe to the selected NWDAF VFL clients(s).

8b. The VFL server sends a Naf_VFLTraining_Unsubscribe to the selected trusted AF VFL clients(s).

8c. For each selected untrusted AF VFL clients, the VFL server sends a Nnef_VFLTraining_AFClient_Unsubscribe to the NEF handling that AF.

8d. For each selected untrusted AF VFL clients, the NEF sends a Naf_VFLTraining_Unsubscribe to that AF.

8e. An NWDAF VFL client aggregator may send Nnwdaf_VFLTraining_Unsubscribe to other one or more indirect NWDAF VFL client in step 2e.

9. The VFL Server, stores VFL correlation ID, the local trained ML Model, the mapping information of the VFL correlation ID to the following parameters: Analytics ID related to the VFL training process, locally trained Model. Additionally, the VFL server stores the VFL client information, which may be used to determine associated VFL client in the VFL inference.

Each VFL client stores VFL correlation ID, the locally trained ML Model, the mapping information of the VFL correlation ID to locally trained Model

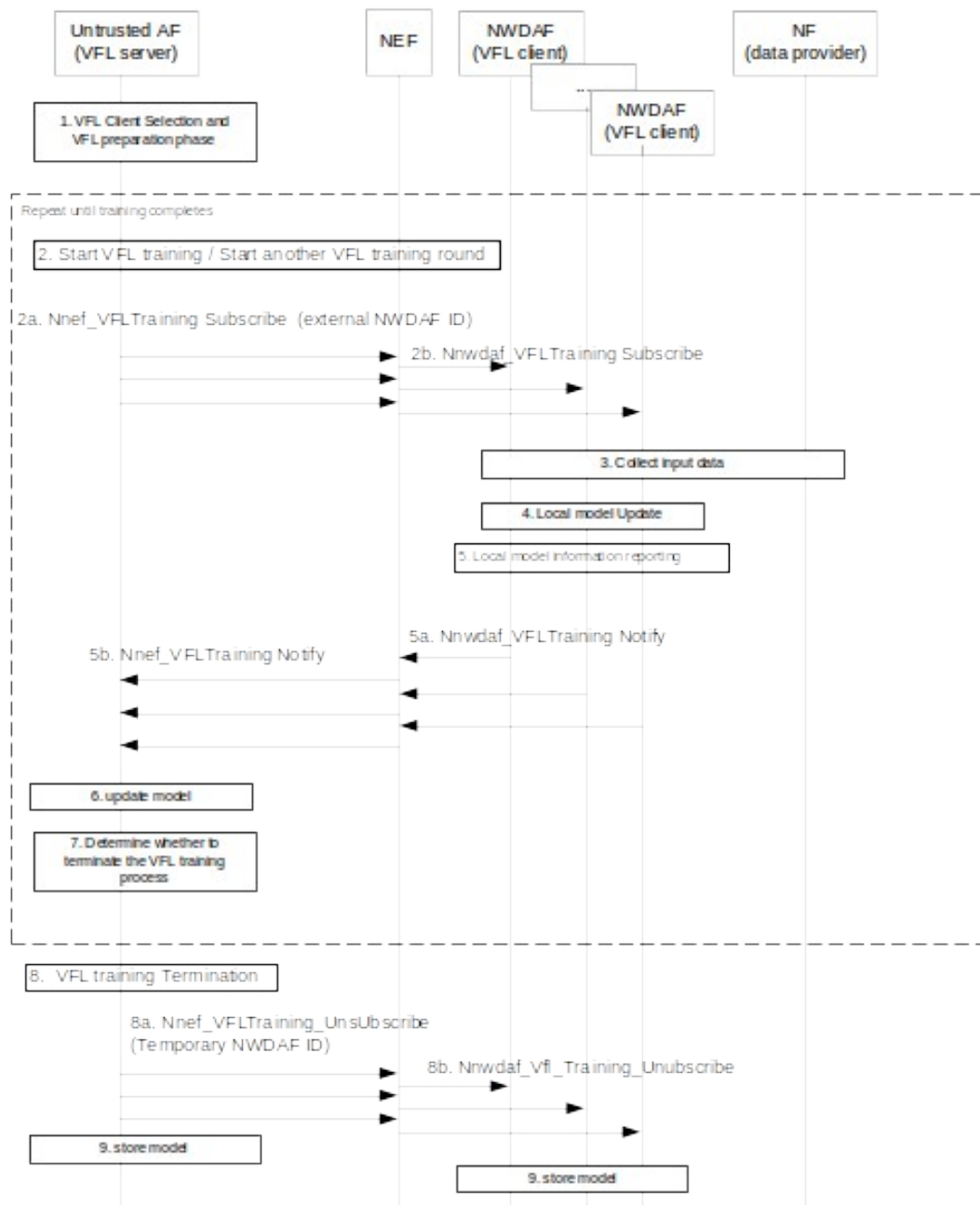
NOTE 4: The VFL correlation ID and the stored mapping information are used later for inference as described in Clause 6.2H.2.4.1.

Editor's note: Whether VFL Training termination Flag in the termination message is required is determined after settling down the service operation.

NOTE 5: If untrusted AF is involved in VFL Clients, the message between NWDAF acting as VFL Server and the untrusted AF is via NEF.

*****Next Change*****

6.2H.2.3.21 Training Procedure for Vertical Federated Learning untrusted AF is acting as VFL server



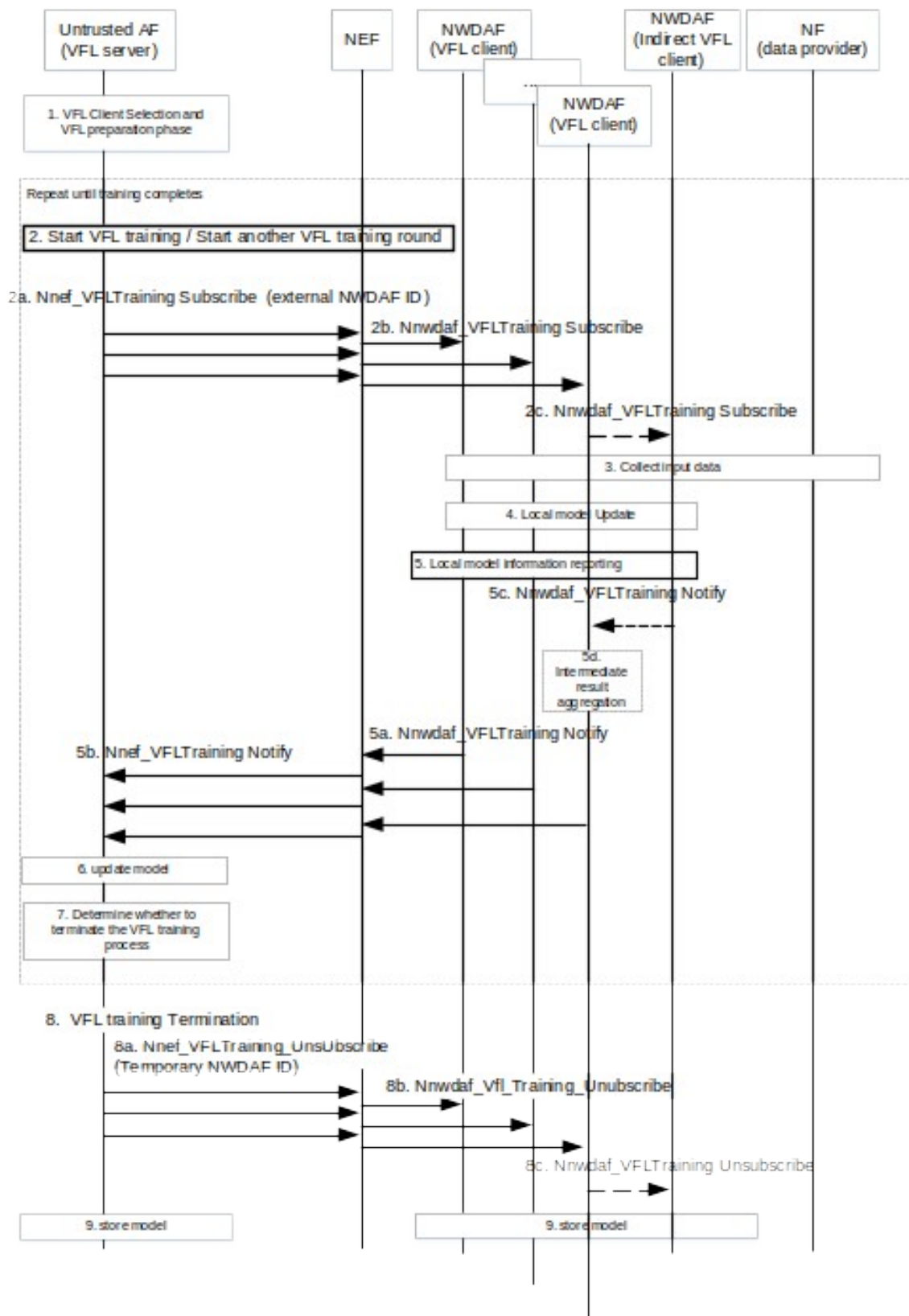


Figure 6.2H.2.3.2-1: Training procedure for Vertical Federated Learning when untrusted AF is acting as VFL server

Editor's note: The ENs listed in clause 6.2H.2.3.1 are also applied to this clause.

1. Same as step 1 in Figure 6.2H.2.3.1-1.

Steps 2-7 are repeated until the training termination condition is reached.

2. To start VFL training, the VFL server do same as in step 4 in Figure 6.2H.2.3.1-1, using Nnef_VFLTraining_Subscribe.

2c. If a NWDAF VFL client is selected as VFL client aggregator by the VFL server, this An-NWDAF VFL client may send Nnwdaf_VFLTraining_Subscribe to other one or more indirect NWDAF VFL client as configured by the VFL server from which it desires to receive the client intermediate training results

~~Steps 3-7 are repeated until the training termination condition is reached.~~

3. [Optional] Same as step 3 in Figure 6.2H.2.3.1-1.

4. Same as step 4 in Figure 6.2H.2.3.1-1.

5. Same as step 5 in Figure 6.2H.2.3.1-1.

5a. A NWDAF VFL client sends a Nnwdaf_VFLTraining_Notify.

5b. For an untrusted AF acting as VFL server, the NEF converts any internal identifiers to external identifiers, provides the external NWDAF ID and sends a Nnef_VFLTrainingNotify to the VFL server.

5c-5d The indirect NWDAF VFL clients share the client intermediate training results with the NWDAF VFL client aggregator from which it received subscription request in step 2c. This NWDAF VFL client aggregates the received client intermediate training results from the indirect NWDAF VFL clients, performs local computation, then sends one Nnef_VFLTraining_Notify notify message to the NEF by including its client intermediate training result.

~~Editor's Note: It is FFS how the client aggregator is selected and may not be selected/configured by VFL server, and also if this capability needs to be registered into NRF.~~

6. [Optional] Same as step 6 in Figure 6.2H.2.3.1-1.

7. Same as step 7 in Figure 6.2H.2.3.1-1.

8. Same as step 8 of Figure 6.2H.2.3.1-1. However, sub steps in that figure are not applicable.

8a. For each NWDAF VFL client, the untrusted AF as VFL server sends a Nnef_VFLTraining Unsubscribe to the NEF handling that AF. The untrusted AF identifies the VFL client using the external NWDAF ID assigned in the discovery procedure (see clause 6.2H.2.1.1).

8b. The NEF sends an Nnwdaf_VFLTraining_Unsubscribe to the NWDAF VFL client indicated by the received external NWDAF ID.

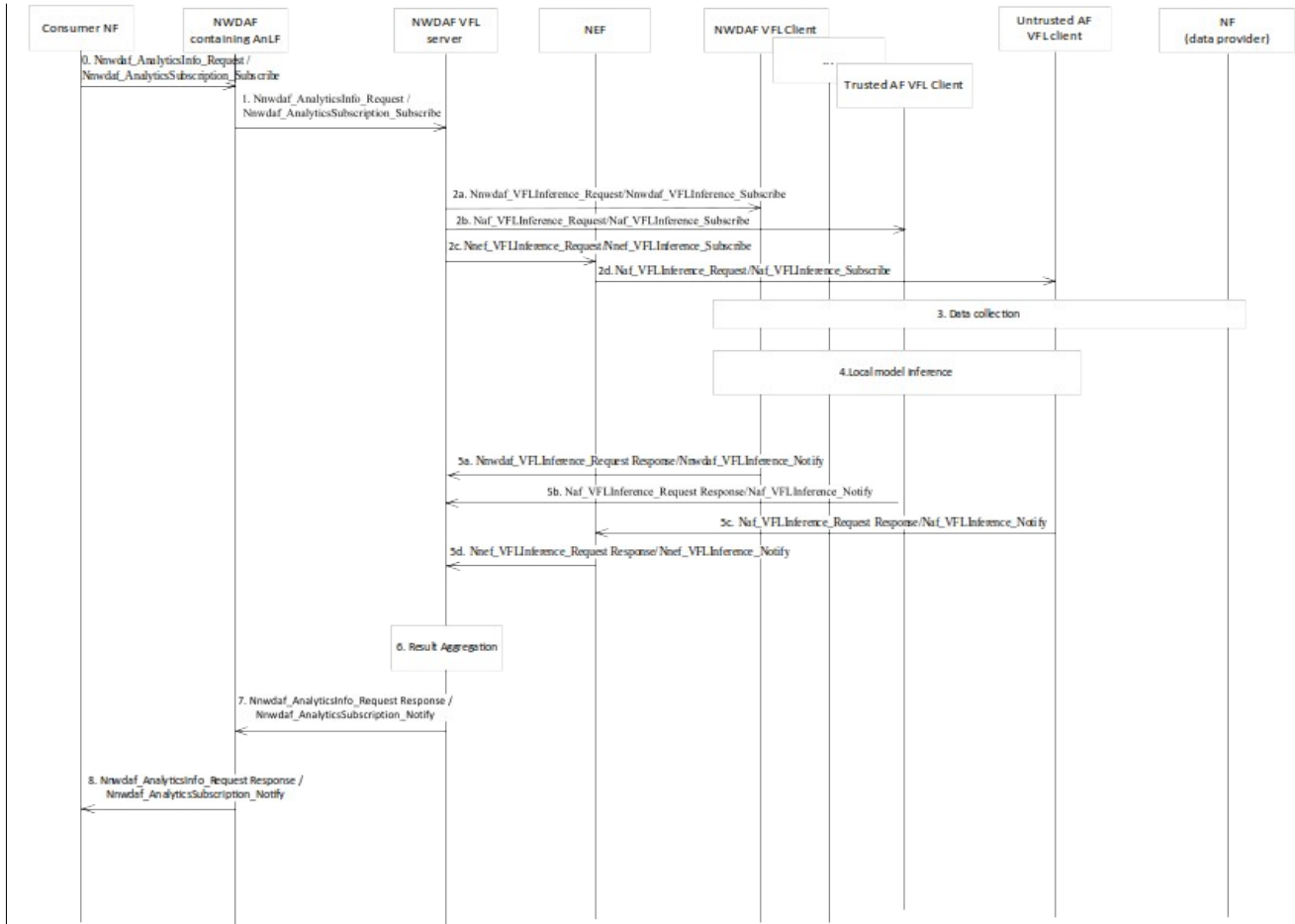
8c. An NWDAF VFL client may send Nnwdaf_VFLTraining_Unsubscribe to other one or more indirect NWDAF VFL client in step 2c.

9. Same as step 9 of Figure 6.2H.2.3.1-1.

*****Next Change*****

6.2H.2.4.1

Inference procedure for vertical federated learning when NWDAF or Trusted AF is acting as VFL server



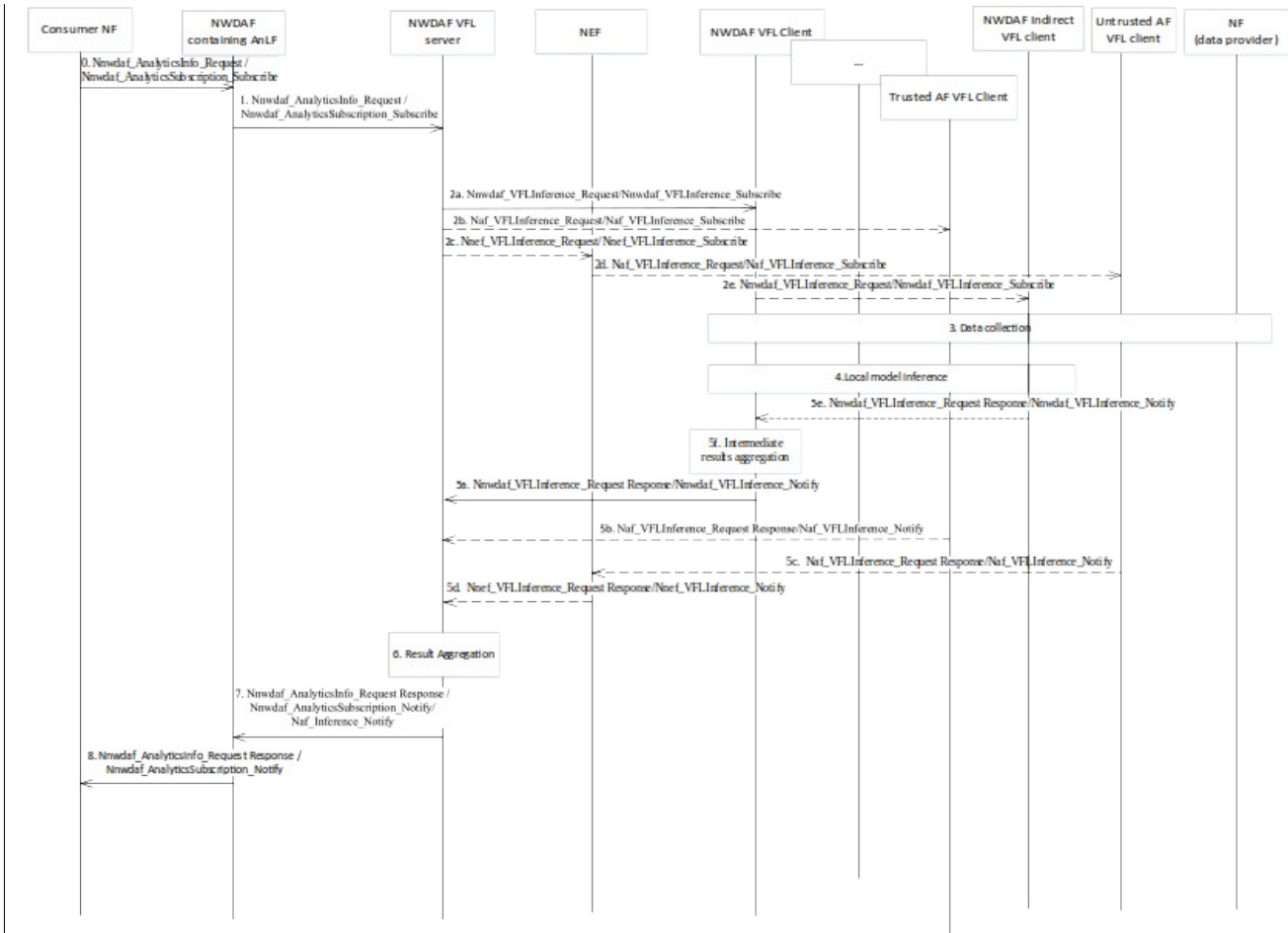


Figure 6.2H.2.4.2-1: Inference procedure for vertical federated learning when NWDAF or Trusted AF is acting as VFL server

0. The analytics consumer NF sends an Analytics request/subscribe (Analytics ID, Target of Analytics Reporting= e.g. UE IDs and optionally Analytics Reporting Information=Analytics target period and Analytics Filter) to NWDAF containing AnLF by invoking a Nnwdaf_AnalyticsInfo_Request or a Nnwdaf_AnalyticsSubscription_Subscribe.

1. If the NWDAF containing AnLF can be the VFL server to generate the VFL inference results for the requested analytics ID, then step 1 is skipped.

If the NWDAF containing AnLF can not generate the analytics output, the NWDAF containing AnLF determines the VFL Server for the requested **analytics**, sends a subscription request to NWDAF VFL server using Nnwdaf_AnalyticsInfo_Request or Nnwdaf_AnalyticsSubscription_Subscribe including Analytics ID, Target of Analytics Reporting = e.g. UE IDs and optionally Analytics Reporting Information=Analytics target period and Analytics Filter.

2. Based on the information received in the step 0 or 1, VFL server decides to initiate the VFL inference procedure with the VFL clients. VFL Server selects clients(s) using information stored in the VFL training process. The server may select some or no clients, e.g. depending on the accuracy of the VFL model, the contribution to the training result and the current status of the VFL clients.

When no VFL Clients are selected, the VFL server may generate the VFL inference results based only on its local trained ML model associated with the determined VFL correlation ID, skipping the steps 2 - 6 (and 7 if step 1 was also skipped).

VFL server NWDAF sends a VFL Inference request/subscription to the VFL clients including the Target of VFL inference = e.g. UE IDs, VFL correlation ID to indicate the VFL client which previously well-trained VFL local model associated with this ID will be used and optionally VFL inference filter.

Editor's note: It is FFS whether additional parameters are needed to send from the VFL server to VFL client, e.g. parameters used in training phase and parameters from Analytics request.

Editor's note: It is FFS how the origin of analytics results can be traced and explained if not all clients participate and results are not satisfying.

2a. For each NWDAF VFL client, the VFL Server NWDAF sends an Nnwdaf_VFLInference_Subscribe or Nnwdaf_VFLInference_Request to the VFL client.

2b. For each trusted AF VFL client, the VFL Server NWDAF sends an Naf_VFLInference_Subscribe or Naf_VFLInference_Request to the VFL client.

2c. For each untrusted AF VFL client, the VFL Server NWDAF sends an Nnef_VFLInference_Subscribe or Nnef_VFLInference_Request to the NEF serving the AF.

2d. For each untrusted AF VFL client, the NEF converts any internal identifiers to external identifiers and sends an Naf_VFLInference_Subscribe or Naf_VFLInference_Request to the untrusted AF VFL client.

2e. An NWDAF VFL client aggregator may send Nnwdaf_VFLInference_Subscribe or Nnwdaf_VFLInference_Request to other one or more indirect NWDAF VFL client(s) as configured by the VFL server from which it desires to receive the client intermediate inference results

3. Each VFL Client collects its local data by using the current mechanism if the VFL Client does not have local data available already.

4. Based on the VFL correlation ID, each VFL Client determines the VFL local model to generate the intermediate local inference results.

~~Editor's note: Whether VFL client may also provide local intermediate inference results to other VFL client is FFS.~~

5. VFL Client sends the client intermediate results to the VFL server.

The intermediate results, which are sent from the VFL Client to the VFL Server during the VFL inference process, are the information for the VFL Server to combine and generate the VFL inference results.

Editor's note: It is FFS additional parameters are needed to send from the VFL client to VFL server.

If the VFL server used an inference subscription in step 2, step 5 may be repeated.

5a. Each NWDAF VFL client sends an Nnwdaf_VFLInference_Notify or Nnwdaf_VFLInference_Request response to the VFL Server NWDAF.

5b. Each trusted AF VFL client sends a Naf_VFLInference_Notify or Naf_VFLInference_Request response to the VFL Server NWDAF.

5c. Each untrusted AF VFL client sends a Naf_VFLInference_Notify or Naf_VFLInference_Request response to the NEF.

5d. For each untrusted AF VFL client, the NEF converts any external to internal identifiers and sends an Nnef_VFLInference_Notify or Nnef_VFLInference_Request response to the NWDAF VFL server.

5e – 5f. The indirect NWDAF VFL clients may share the client intermediate inference results with a NWDAF VFL client aggregator from which it received request in step 2e. Theis NWDAF VFL client aggregator aggregates the received client intermediate inference results, performs local computation, then sends one Nnwdaf_VFLInference_Notify or Nnwdaf_VFLInference_Request response that includes the client intermediate inference result of theis NWDAF VFL client aggregator to the VFL server.

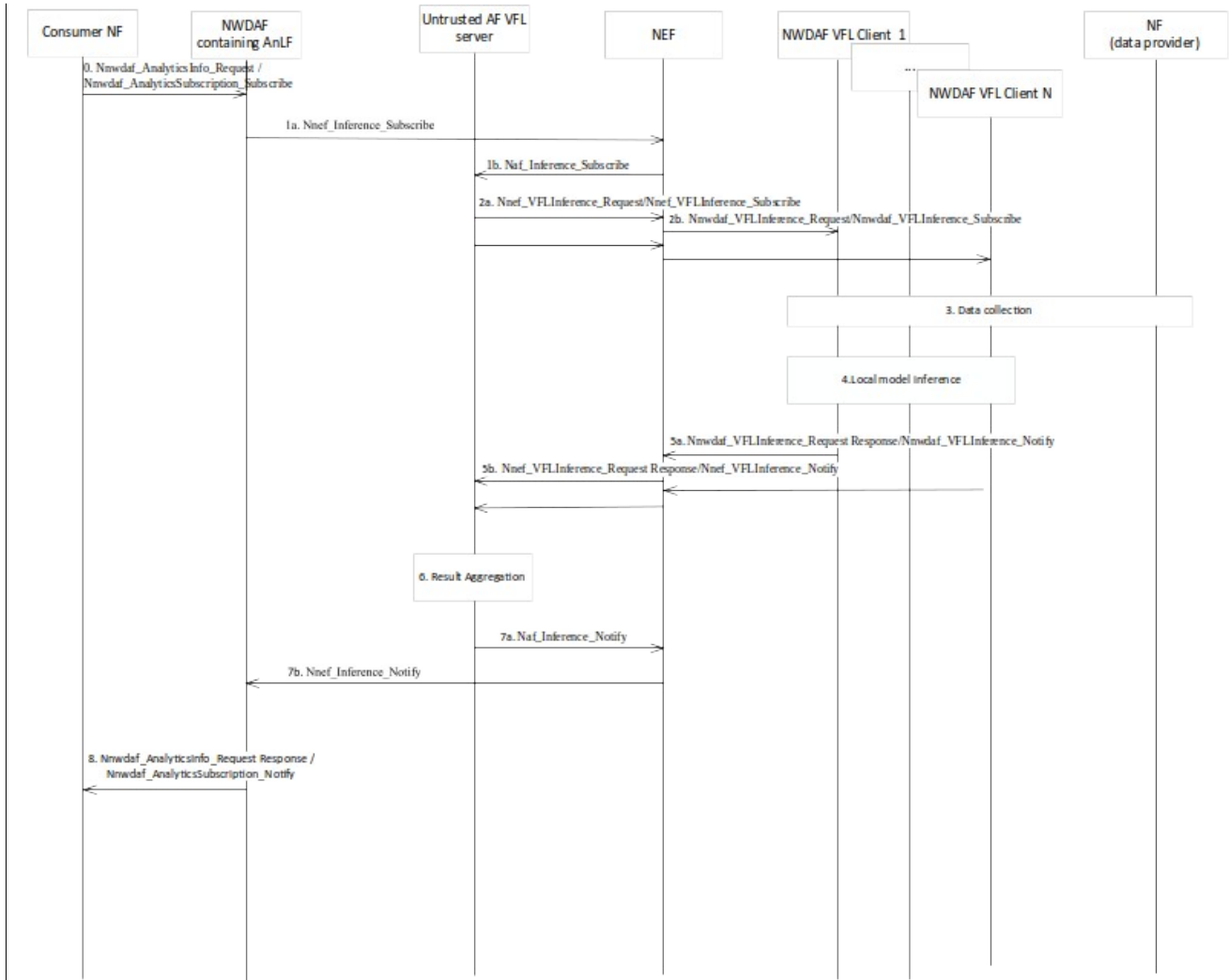
6. The VFL server may collect its local data and generate the intermediate local inference results. When the VFL Server selected VFL clients to participate in the VFL Inference process, it combines all the intermediate results to generate the VFL inference results based on the VFL correlation ID. The VFL server takes into account the participation of each VFL client during the ML training process and the importance of the intermediate results when generates the combined inference output.

7. Depending on request, the NWDAF VFL server sends Nnwdaf_AnalyticsInfo_Response or Nnwdaf_AnalyticsSubscription_Notify to the consumer (i.e NWDAF containing AnLF) including the VFL inference results.

8. The NWDAF containing AnLF provides the analytics output to the analytics consumer NF based on the VFL inference results by means of either Nnwdaf_AnalyticsInfo_Response or Nnwdaf_AnalyticsSubscription_Notify, depending on the service used in step 0.

6.2H.2.4.2

Inference procedure for vertical federated learning when untrusted AF is acting as VFL server



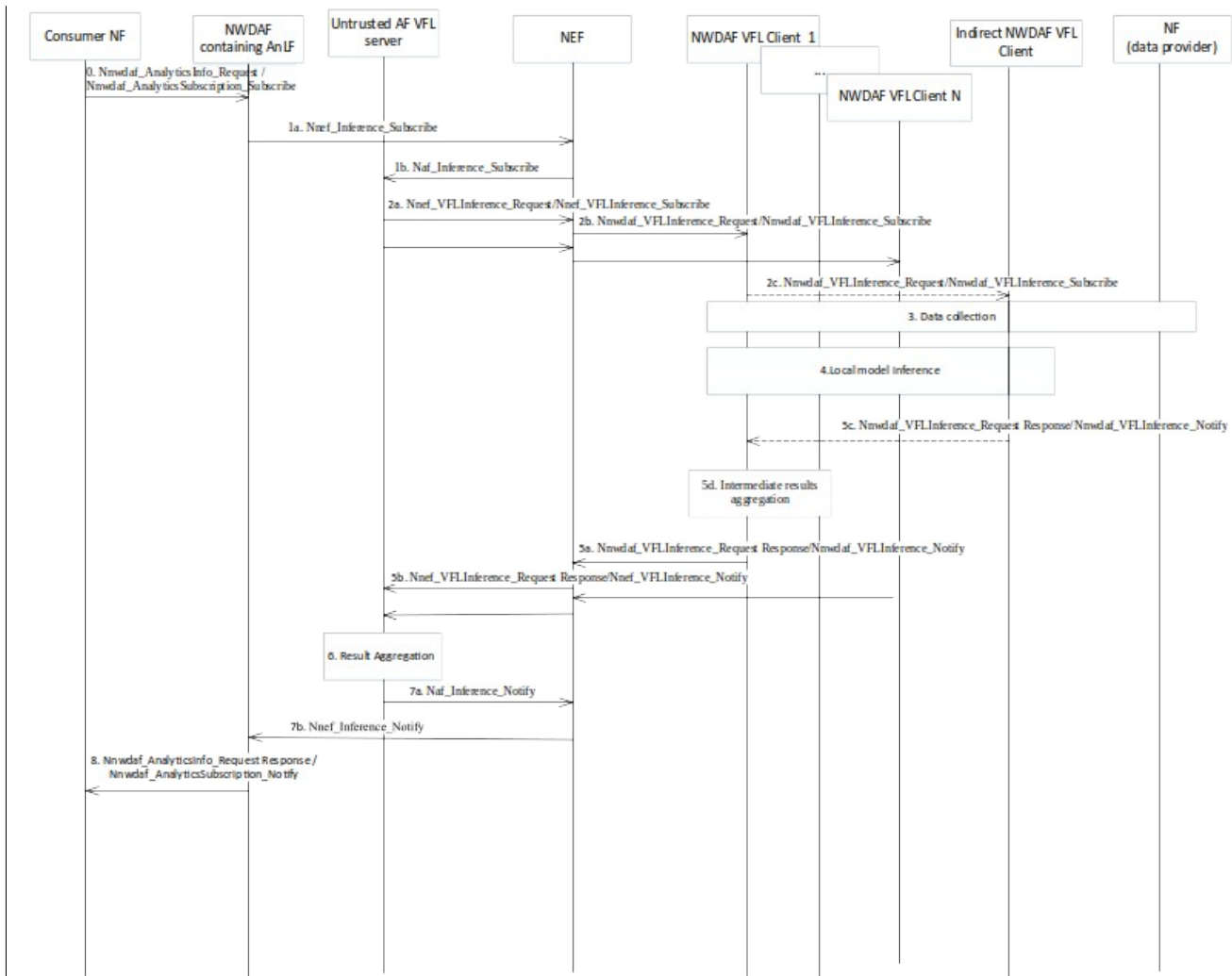


Figure 6.2H.2.4.2-1: Inference procedure for vertical federated learning when untrusted AF is acting as VFL server

The inference procedure when untrusted AF is acting as VFL server may be triggered by a request or subscription from a 5GC consumer NF or internal service logic of the AF acting as VFL server. If triggered by internal service logic of the AF acting as VFL server, the steps 0, 1, 7 and 8 are skipped. The inference procedure triggered by internal service logic of the AF acting as VFL server is out of 3GPP scope.

0. Same as step 0 in clause 6.2H.2.4.1.

1. Same as step 1 in clause 6.2H.2.4.1, to NEF using Nnef_Inference_subscribe. NEF forwards the subscription request to AF using Naf_Inference_subscribe.

Editor's note: When the AnLF determine the VFL server AF is FFS. For example, during the training phase.

Editor's note: The service name between the AnLF and untrusted AF as VFL server is FFS.

2. Same as step 2 in clause 6.2H.2.4.1, to NEF using Nnef_VFLInference_subscribe. An untrusted AF includes the external NWDAF ID and sends the request to the NEF.

NEF converts any received external identifiers to internal identifiers and forwards the subscription request to NWDAF using Nnwdaf_VFLInference_subscribe.

2c. An NWDAF VFL client aggregator may send Nnwdaf_VFLInference_Subscribe or Nnwdaf_VFLInference_Request to the VFL client to other one or more indirect NWDAF VFL client as configured by the VFL server from which it desires to receive the client intermediate inference results

Editor's note: It is FFS additional parameters are needed to send from the VFL server to VFL client.

3. Same as step 3 in clause 6.2H.2.4.1.
4. Same as step 4 in clause 6.2H.2.4.1.

Editor's note: Whether VFL client may also provide local intermediate inference results to other VFL client is FFS.

5. Same as step 5 in clause 6.2H.2.4.1, to NEF using Nnwdaf_VFLInference_Notify.

NEF converts any received internal identifiers to external identifiers and forwards the subscription notify to the untrusted AF using Nnef_VFLInference_Notify.

5c-5d. The indirect NWDAF VFL clients share the client intermediate inference results with the NWDAF VFL client aggregator from which it received request in step 2c. This NWDAF VFL client aggregator aggregates the received client intermediate inference results, performs local computation then sends one notify message to the NEF by including its client intermediate inference result.

Editor's note: It is FFS additional parameters are needed to send from the VFL client to VFL server.

If the VFL server used an inference subscription in step 2, step 5 may be repeated.

6. Same as step 6 in clause 6.2H.2.4.1.
7. Same as step 7 in clause 6.2H.2.4.1, to NEF using Naf_Inference_Notify.

NEF converts any received internal identifiers to external identifiers and forwards the subscription Notify to NWDAF using Nnef_Inference_Notify.

8. Same as step 8 in clause 6.2H.2.4.1.

*****End of Changes*****